

OTARU, Japan

WMO: 474110

Lat: 43.1825N

Lon: 141.0170E

Elev: 26

StdP: 101.01

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.5	-8.3	-15.1	1.0	-8.3	-13.6	1.2	-7.0	9.3	-1.3	8.4	-2.5	4.0	250	0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.4	28.4	22.4	26.8	21.6	25.4	20.8	23.4	27.1	22.6	25.7	21.8	24.5	3.0	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	17.0	25.4	21.5	16.2	24.8	20.7	15.4	24.1	70.2	27.2	67.1	25.7	64.0	24.6	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.0	6.9	6.1	-11.4	31.8	1.7	1.4	-12.6	32.8	-13.6	33.6	-14.6	34.4	-15.8	35.5
			-12.2	24.4	1.6	1.1	-13.4	25.3	-14.3	25.9	-15.2	26.6	-16.4	27.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-3.4	-2.8	0.7	6.6	12.3	16.2	20.5	22.0	18.3	11.9	4.8	-1.1
	DBStd	9.39	2.73	3.00	3.14	3.38	3.26	2.77	2.49	2.39	2.81	3.16	4.06	3.13
	HDD10.0	1711	414	359	288	112	14	0	0	0	0	18	161	346
	HDD18.3	3678	673	592	546	351	188	74	8	1	34	201	406	604
	CDD10.0	1311	0	0	0	11	86	187	325	372	249	76	5	0
	CDD18.3	236	0	0	0	0	1	11	74	116	34	0	0	0
	CDH23.3	902	0	0	0	1	17	36	240	496	112	0	0	0
	CDH26.7	154	0	0	0	0	2	4	42	91	16	0	0	0
Wind	WSAvg	2.7	3.3	3.2	3.1	2.8	2.4	2.0	1.8	1.9	2.3	2.8	3.1	3.3
Precipitation	PrecAvg	1403	124	103	88	84	79	106	157	129	161	109	127	136
	PrecMax	1608	184	201	148	127	108	152	237	245	299	184	225	222
	PrecMin	1072	63	57	39	38	46	74	104	58	60	52	66	84
	PrecStd	139	31	33	29	26	18	20	40	53	70	33	41	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.4	7.0	11.4	21.1	25.6	26.8	30.2	31.5	28.8	21.7	16.9	9.9
		MCWB	2.7	3.9	6.5	10.9	14.8	18.4	23.4	23.8	22.5	15.5	11.6	7.1
	2%	DB	3.2	4.5	8.9	16.7	22.3	24.0	27.7	28.9	26.1	19.7	14.2	6.6
		MCWB	1.1	1.8	4.9	10.1	14.1	17.7	22.3	23.0	20.6	14.6	10.2	4.2
	5%	DB	1.9	3.1	7.0	14.1	19.7	22.3	25.9	27.3	24.3	18.3	12.6	5.0
		MCWB	0.0	0.8	3.7	8.5	13.0	17.0	21.2	22.3	19.5	13.9	9.0	2.8
	10%	DB	0.7	1.9	5.5	12.0	17.6	20.8	24.3	26.0	22.9	16.8	10.9	3.4
		MCWB	-1.2	-0.3	2.6	7.5	12.2	16.5	20.5	21.6	18.6	12.7	7.9	1.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	4.1	7.0	12.8	16.7	20.0	24.2	24.6	23.2	17.0	12.5	7.4
		MCDB	4.9	6.5	10.5	18.9	22.9	23.8	28.7	29.1	27.2	19.3	15.6	9.8
	2%	WB	1.4	2.2	5.4	10.5	15.0	18.7	23.0	23.8	21.8	15.6	10.7	4.6
		MCDB	2.9	4.1	8.3	15.4	20.2	22.6	26.5	27.5	25.1	18.5	13.6	6.5
	5%	WB	0.2	1.0	4.1	9.2	13.9	17.7	22.1	23.1	20.5	14.4	9.4	3.0
		MCDB	1.7	2.9	6.7	13.4	18.3	21.3	24.7	26.3	23.4	17.5	12.1	4.8
	10%	WB	-1.0	-0.1	2.8	7.9	12.9	16.9	21.2	22.3	19.3	13.3	8.0	1.4
		MCDB	0.5	1.7	5.2	11.6	16.7	20.0	23.7	25.2	22.1	16.3	10.7	3.2
Mean Daily Temperature Range	5% DB	MDBR	4.7	5.2	5.8	7.5	8.2	7.1	6.2	6.4	7.2	7.2	6.0	4.7
		MCDBR	6.5	7.4	8.6	11.5	12.3	9.9	8.6	8.3	8.5	8.9	8.8	7.2
		MCWBR	5.5	6.1	6.0	6.6	6.3	5.0	4.2	4.2	5.0	5.7	6.5	5.8
	5% WB	MCDBR	6.5	7.3	8.4	10.8	10.7	8.5	7.5	7.2	7.5	8.1	8.6	7.2
		MCWBR	5.7	6.3	6.2	6.6	6.0	4.6	3.9	3.8	4.8	5.6	6.8	6.1
Clear-Sky Solar Irradiance	taub	0.350	0.399	0.464	0.524	0.539	0.522	0.504	0.469	0.415	0.409	0.393	0.367	
	taud	2.025	1.890	1.799	1.788	1.852	1.981	2.101	2.218	2.317	2.182	2.121	2.022	
	Ebn at Noon	733	755	756	748	751	767	776	792	810	750	684	666	
	Edh at Noon	112	152	190	208	200	177	155	134	113	113	102	103	
All-Sky Solar Radiation	RadAvg	1.16	1.89	3.20	4.50	5.12	5.00	4.54	4.39	3.73	2.60	1.42	0.96	
	RadStd	0.09	0.18	0.29	0.46	0.47	0.48	0.50	0.34	0.35	0.16	0.15	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page